

Homework 4

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Assignment

Let $f_n : (0, \infty) \rightarrow \mathbb{R}$ be given by

$$f_n(x) = \frac{n}{nx + 1}.$$

Prove that the sequence (f_n) converges uniformly on every set of the form $[a, \infty)$, but that the convergence is not uniform on $(0, \infty)$.

Proof

We first verify *pointwise convergence*. Fix any $x > 0$. Then

$$f_n(x) = \frac{n}{nx + 1} = \frac{1}{x + \frac{1}{n}},$$

and this expression clearly tends to $\frac{1}{x}$ as $n \rightarrow \infty$. Hence

$$\lim_{n \rightarrow \infty} f_n(x) = \frac{1}{x}.$$

Next, we show that the convergence cannot be *uniform on all of* $(0, \infty)$. For uniform convergence, we would require

$$\lim_{x > 0} (\sup |f_n(x) - f(x)|) = 0.$$

A more convenient form of the difference is

$$|f_n(x) - f(x)| = \left| \frac{n}{nx + 1} - \frac{1}{x} \right| = \left| \frac{nx - (nx + 1)}{x(nx + 1)} \right| = \frac{1}{nx^2 + x}.$$

If we let $x = \frac{1}{\sqrt{n}}$, then

$$nx^2 + x = n \cdot \frac{1}{n} + \frac{1}{\sqrt{n}} = 1 + \frac{1}{\sqrt{n}},$$

so

$$\frac{1}{nx^2 + x} = \frac{1}{1 + \frac{1}{\sqrt{n}}} \approx 1 \quad \text{for large } n.$$

This shows that

$$\sup_{x>0} \frac{1}{nx^2 + x} \geq \frac{1}{1 + \frac{1}{\sqrt{n}}},$$

which does not tend to 0 as $n \rightarrow \infty$. Therefore, the convergence is not uniform on $(0, \infty)$.

Finally, we verify that the convergence *is uniform on each* $[a, \infty)$ for any fixed $a > 0$. Indeed, if $x \geq a > 0$, then

$$nx^2 + x \geq na^2 + a, \quad \text{so} \quad |f_n(x) - f(x)| = \frac{1}{nx^2 + x} \leq \frac{1}{na^2 + a}.$$

Hence,

$$\sup_{x \in [a, \infty)} |f_n(x) - f(x)| \leq \frac{1}{na^2 + a},$$

and the right-hand side $\frac{1}{na^2 + a}$ tends to 0 as $n \rightarrow \infty$, independent of x . Thus, given any $\varepsilon > 0$, we can choose N large enough so that $n \geq N$ implies

$$\frac{1}{na^2 + a} < \varepsilon,$$

and consequently

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for all } x \in [a, \infty).$$

This completes the proof that f_n converges uniformly to $\frac{1}{x}$ on each $[a, \infty)$, but not on $(0, \infty)$ as a whole.

References

I came up with the proof by myself. Then, I used ChatGPT-o1 to convert my handwritten proof into a LaTeX document for readability.